CDT Healthcare Data Science

ML for Time Series Module

Practical 04: End-to-End and Foundation Models for ECG

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0 Lab overview

In this lab we learn two ways to classify ECGs without hand-crafted features: training a deep network end to end, and using a pretrained foundation model. We compare them with the random forest of Lab 1 on three tasks: atrial fibrillation (AF), gender, and sleep apnea.

The three methods on three tasks give nine baseline scores on hidden test sets (Table 1). We end the lab by trying to beat them. We will evaluate your implementation of the four functions and how far you beat the baselines.

System	Validation				Hidden test			
	AF	**Sex**	**Apnea**	**Geo. mean**	**AF**	**Sex**	**Apnea**	**Geo. mean**
Random Forest (Lab 1)	0.619	0.684	0.900	0.725	0.591	0.614	0.644	0.616
ResNet (Section 2)	0.608	0.662	0.832	0.694	0.555	0.600	0.514	0.555
ECGFounder + linear classifier	0.635	0.699	0.932	0.745	0.655	0.679	0.613	0.648

Table 1: The baselines: the three systems of this lab on the validation splits and on the hidden test sets.

How to work. Open `lab2-end-to-end.ipynb` and run the cells in order. Each task asks you to complete one function, described in a box in this note. When you run it, the notebook checks your answer against a reference. If you have not finished a task, the notebook uses the reference instead, so the rest of the cells still run.

Setup.

1. Open the notebook in Colab using the badge in its first cell.
2. Save a copy to your Drive, using *Save a copy in Drive* in the File menu.
3. Choose a GPU: *Runtime > Change runtime type > T4 GPU*. The whole notebook takes about ten minutes on the free T4.
4. Run the Setup cell. It downloads the code, the data and the ECGFounder weights.
5. Enter the data password when asked. Your tutor will give it out during the lab.

1 Preparing the three tasks

Notebook: Section 1

We set up the benchmark in this section: three classification tasks (Table 2) and their metrics. Every input is one ECG lead at 300 Hz. We have preprocessed and split the data. You get the raw signals and the labels of the training and validation splits, and only the raw signals of the test sets. We score a model with the metric of its task (Table 3), and we evaluate the system by the geometric mean of its three task scores.

Task	Train	Validation	Hidden test	Metric
AF: `N`, `A`, `O`, `~`	1,500 PhysioNet 2017 recordings [1, 2]	500	6,528 other PhysioNet 2017 recordings	macro F1
Sex: female or male	3,258 PTB-XL ECGs [3]	1,123	1,000 Chapman–Shaoxing ECGs [4]	AUROC
Apnea: yes or no	5,676 Apnea-ECG minutes [5]	1,939	3,471 UCD St Vincent's minutes [6]	AUROC

Table 2: Details of AF, Sex and Apnea prediction task. Every system is trained on the training split and evaluated on the validation split.

Metric	Used for	Definition
Macro F1	AF	$\frac{1}{4}\sum_c \frac{2P_cR_c}{P_c+R_c}$
AUROC	Sex, apnea	$\frac{1}{n_+n_-}\sum_{i\in+}\sum_{j\in-}\mathbf{1}[s_i > s_j]$
Geo. Mean	System comparison	$\sqrt[3]{\text{AF} \times \text{Sex} \times \text{Apnea}}$

Table 3: The metrics of the benchmark.

2 End-to-end classification: ResNet

Notebook: Section 2, Tasks 2A and 2B

We train a ResNet end to end on the raw signal to predict AF in this section. The network reads a 10 s crop of the cleaned signal and returns the probabilities of the four AF classes. We build the pipeline from four functions (Figure 1), and we reuse it for the sex and apnea tasks.

Task 2A. Write `crop`. *Submit:* your implementation of the function.

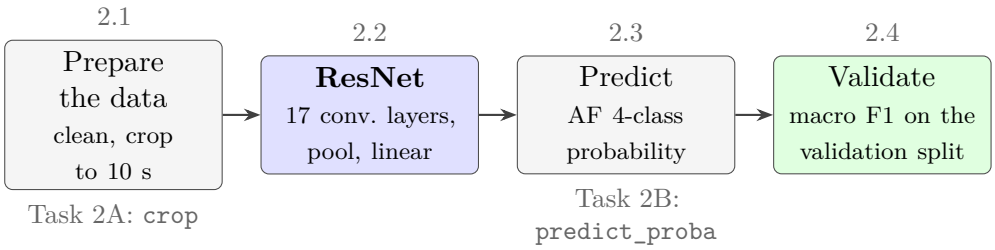


Figure 1: The ResNet pipeline. The numbers name the subsection that builds each part; you write the two functions named below the boxes. The validation score picks the best epoch.

Specification: `crop(x, length, train)`

Given `x`, a float array of shape `(T,)`; `length`, an int; `train`, a bool.

You may use `np.random.randint`, `np.pad`.

Return a float array of shape `(length,)`.

Task 2B. Write `predict_proba`. *Submit:* your implementation of the function.

Specification: `predict_proba(model, signals, length, batch_size=256)`

Given `model`, a `resnet1d18` on `device`; `signals`, a list of n float arrays of shape `(T_i,)`; `length`, an int; `batch_size`, an int.

You may use `make_batch` (2.1), `model.eval()`, `torch.no_grad()`, `torch.softmax`, `np.concatenate`.

Return a float array of shape `(n, n_classes)`, class probabilities.

3 Foundation model classification: ECGFounder

Notebook: Section 3, Tasks 3A and 3B

We trained a ResNet from scratch in the last section and saw it fall below the random forest. One explanation is that the features a ResNet learns from 1,500 recordings are poorer than the hand-crafted ones [1]. A foundation model combines the strengths of the two: learned features with good performance, because they were learned from far more data. In this section we use ECGFounder [7], trained on 10.8 million hospital ECGs, as a feature extractor. It maps each 10 s window to a 1,024-d embedding, and we train a linear classifier on that embedding (Figure 2).

Task 3A. Write `pool_windows`. *Submit:* your implementation of the function.

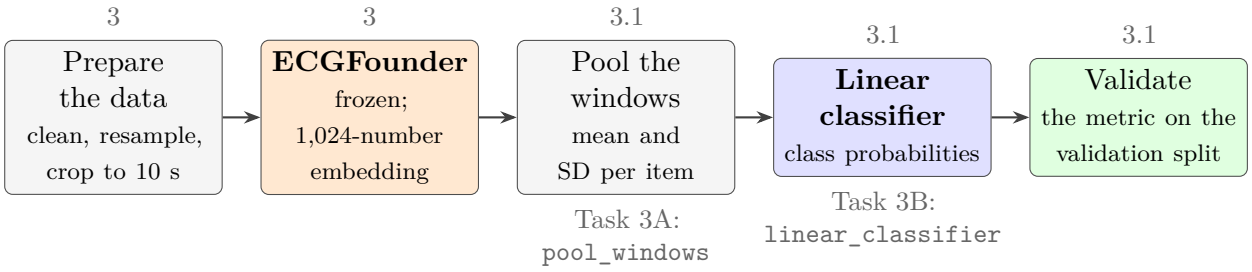


Figure 2: The foundation-model pipeline. The numbers name the subsection that builds each part; you write the two functions named below the boxes. The orange box was trained elsewhere and stays frozen; only the linear classifier is trained on our labels.

Specification: `pool_windows(E, windows)`

Given `E`, a float array of shape `(n * windows, d)`; `windows`, an int.

You may use `np.reshape`, `np.mean`, `np.std`, `np.concatenate`.

Return a float array of shape `(n, 2d)`.

Task 3B. Write `linear_classifier`. *Submit:* your implementation of the function.

Specification: `linear_classifier(E_train, y_train, E_eval)`

Given `E_train`, a float array of shape `(n_train, d)`; `y_train`, an int array of shape `(n_train,)`; `E_eval`, a float array of shape `(n_eval, d)`.

You may use `StandardScaler` and `LogisticRegression` from scikit-learn.

Return a float array of shape `(n_eval, n_classes)`, class probabilities.

4 Beating the baselines

Notebook: Section 4

We now try to beat the three baseline systems. Below are four ideas that could help us.

- **model ensembling**: average the probabilities of two prediction models.
- **feature engineering**: give a classifier the 54 features and the embedding together;
- **hyperparameter tuning**: more epochs, another learning rate, another crop length for the ResNet
- **fine-tuning the foundation model**: train ECGFounder end to end on our labels.

Task 4. Put what we have learned into practice and beat the baselines. *Submit:* your best predictions on the three test sets. The submission system will be evaluated by its performance on the hidden test set. We provide only the label for the validation set, so you have to identify when to stop and when to keep optimizing based on the validation score.

System	Validation				Hidden test			
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ECGFounder + linear classifier	0.635	0.699	0.932	0.745	0.655	0.679	0.613	**0.648**
Instructor's best model	0.788	0.736	0.966	0.824	0.743	0.696	0.652	**0.696**

Table 4: This is the same table as Table 1, with the instructor's best model added

Submitting your work

Submit two things.

1. The notebook with the four functions implemented.
2. The submission zip the notebook writes, with your note of what you tried.

Acknowledgement. Sections 0 to 2 follow the Wearable Lab of the MSc in Applied Digital Health by Dr Xiao Gu. ECGFounder is by Li et al. (PKUDigitalHealth), MIT licence. The sex and apnea data derive from PTB-XL and Chapman–Shaoxing (CC-BY 4.0) and from Apnea-ECG and UCD St Vincent's (ODC-BY 1.0).

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